Chapter 8

Quantification and Analysis of Proteins

Chapter Outline

PROTEIN QUANTITATION

Various platforms and methods are available to quantitate proteins; however, spectrophotometric assays of protein in solution do not require either enzymatic/chemical digestion or separation of the mixture prior to analysis. There are three major types of spectrophotometric assays: UV absorbance methods, dye-binding assays using colorimetric and fluorescent-based detection. Although these spectrophotometric assays can be run at high throughput, they require an appropriate protein standard or constituent amino acid sequence information to make an estimate of concentration. The choice of method used to determine the concentration of a protein or peptide in solution depends on many factors. For practical procedures, the flowchart in Fig. 8.1 describes the process of selecting the most appropriate assay.

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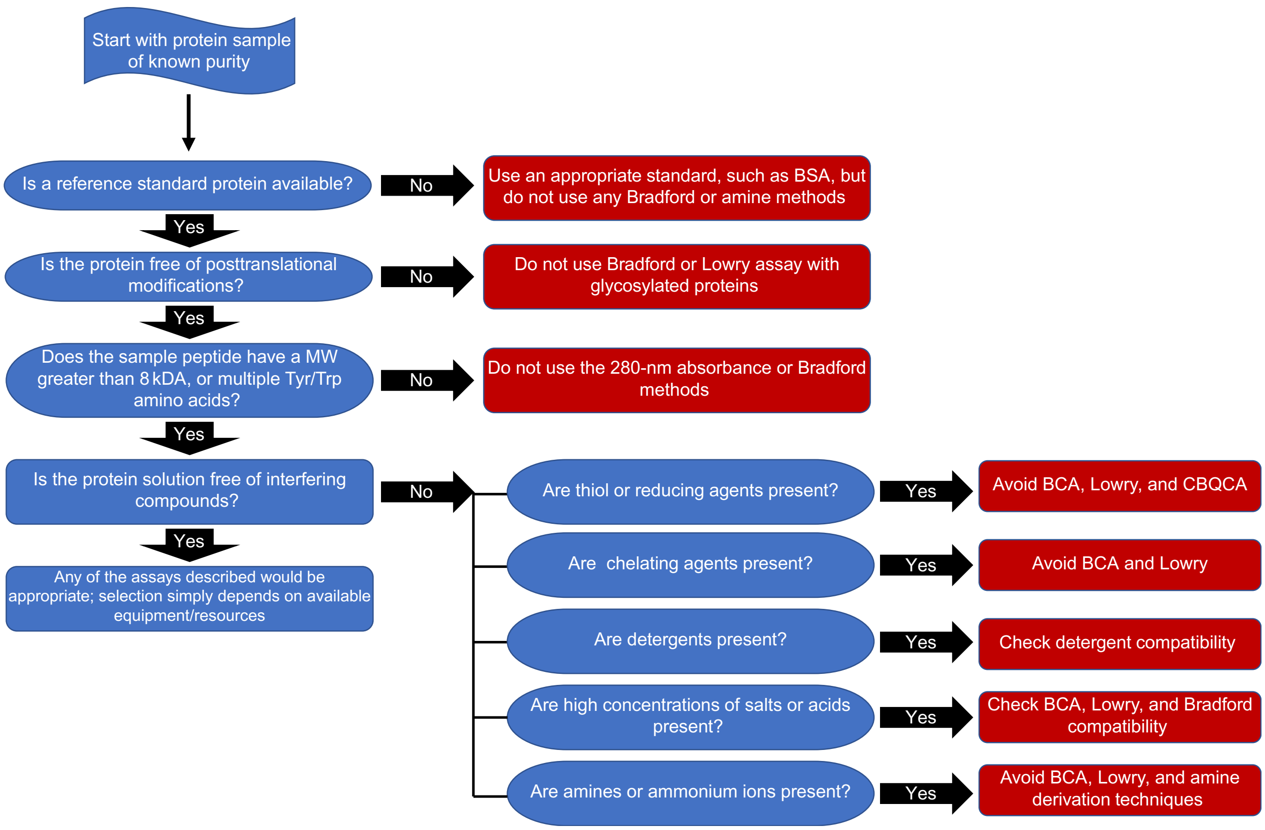


FIG. 8.1 Flow chart for selecting proper techniques for protein quantification upon purification. The chart assumes that the sample for analysis is relatively pure, that is the analyte for quantitation is the major component, for example fractions from affinity chromatography, or extraction from inclusion bodies. The "reference standard protein" refers to a standard that is the same protein that is being quantitated in the same, or similar matrix that is "matched."

Some criteria that need to be considered when selecting an assay include:

- Sample volume: The amount of material available to analyze. Typically, fluorescence-based assays provide the best sensitivity and dynamic range (Fig. 8.2). Microplate assays use lower assay volumes but show improved sensitivity, typically up to 10-fold when compared with cuvette-based assays.
- Sample recovery: If the sample is limited, a nondestructive method such as UV spectroscopy may be more appropriate.
- Throughput: If multiple samples are to be analyzed, a microplate-compatible rapid one-step assay should be considered.
- Robustness: The absorbance-based dye-binding assays appear to display enhanced repeatability and robustness when compared to fluorescent assays.
- Chemical modification: Covalent modification, for example glycosylation or PEGylation, can interfere with specific assays.
- Protein aggregation: The solubility of a protein in solution, often a problem for membrane proteins, or proteins prone to aggregation can alter the expected response for many assays.

UV Spectroscopy Method: Detecting Proteins With Absorbance at 280 nm

At 280 nm, amino acid residues with aromatic rings in a protein absorb light. The amino acids tryptophan (λ_{max} 279.8 nm) and tyrosine (λ_{max} 274.6 nm) have extinction coefficients, ε, of 5.6 and 1.42 $M^{-1}cm^{-1}$, respectively, and thus contribute most to the absorbance at this wavelength, while phenylalanine (λ_{max} 279 nm, ε 0.197 $M^{-1}cm^{-1}$) makes a minor contribution (Table 8.1). The method follows the Beer-Lambert law in which absorbance is proportional to concentration and path length. It should be noted that the method depends on the relative number of aromatic amino acids, primarily tryptophan and tyrosine residues, present in a protein. The method can be used to detect protein in the 20–3000 μg range.

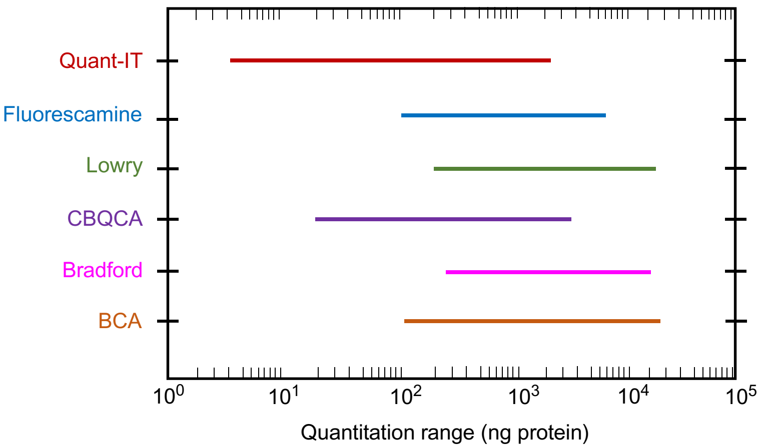


FIG. 8.2 The upper and lower limits for the quantitation range of dye-based protein assay performed in a microplate format.

TABLE 8.1 Wavelength Maxima and Molar Absorptivity (ε) of Select Amino Acids

Amino Acid	Wavelength Maxima (nm)	$\varepsilon \times 10^{-3}$ ($M^{-1} cm^{-1}$)
Cysteine	250	0.3
Histidine	211	5.9
Phenylalanine	188	60.0
	206	9.3
	257	0.2
Tryptophan	219	47.0
	279	5.6
Tyrosine	193	48.0
	222	8.0
	275	1.4

The method is particularly useful for the detection of proteins eluting from chromatography columns, as there is no loss of protein. The possibility of nucleic acid contamination should be considered, and compensation can be made.

Advantages of using absorbance at 280 nm are that this method is fast, easily automated, and reasonably sensitive. Furthermore, this method does not destroy protein. However, many buffers and other reagents can interfere with A_{280} spectrophotometric measurements. The concentration limits for those reagents are listed in Table 8.2. The A_{280} measurement is also interfered with by the absorbance from DNA and RNA. DNA and RNA have an absorbance maximum at 260 nm but still absorb at 280 nm and have tenfold higher absorbance values at 280 nm compared to the equivalent concentration of protein. Therefore, if the sample is contaminated with nucleic acids, the absorbance at 280 nm might not adequately reflect the real protein concentration. To resolve the protein concentration in such samples, measure the absorbance at 260 nm and 280 nm and calculate the protein concentration as follows:

$$\text{Protein concentration (mg/mL)} = 1.55 \times A_{280} - 0.76 \times A_{260}.$$

This estimation of protein concentration is valid up to 20% (w/v) nucleic acid or an A_{280}/A_{260} ratio less than 0.6. The UV absorbance at 280 nm is also only usable for concentration determination if the sequence of the protein of interest contains a known amount of tryptophans and tyrosines. If this is not the case, an alternative is to use Fourier transform infrared

TABLE 8.2 Protein Quantification Limits of Interfering Reagents

Reagent	A_{205}	A_{280}
Ammonium sulfate	9% (w/v)	>50% (w/v)
Brij 35	1% (v/v)	1% (v/v)
DTT (dithiothreitol)	0.1 mM	3 mM
EDTA (ethylenediaminetetraacetic acid)	0.2 mM	30 mM
Glycerol	5% (v/v)	40% (v/v)
KCl	50 mM	100 mM
2-ME (2-mercaptoethanol)	<10 mM	10 mM
NaCl	0.6 M	>1 M
NaOH	25 mM	>1 M
Phosphate buffer	50 mM	1 M
SDS	0.10% (w/v)	0.10% (w/v)
Sucrose	0.5 M	2 M
Tris buffer	40 mM	0.5 M
Triton X-100	<0.01% (v/v)	0.02% (v/v)
TCA (trichloroacetic acid)	<1% (w/v)	10% (w/v)
Urea	<0.1 M	>1 M

spectroscopy (FTIR). After subtracting the contribution of water between 1700 nm and 2300 nm, the analysis of the amide band I and II of the IR absorbance spectrum can be used to calculate protein concentration by determining the concentration of amine bonds. The only limitations of FTIR are the minimal and maximal concentrations that can be used (0.2–5 mg/mL) and the incompatibility of several amine-containing buffers (HEPES $\geq$ 25 mM, Tris $\geq$ 50 mM) or additives (EDTA $\geq$ 10 mM).

Another alternative is amino acid analysis (AAA), which is a valuable technique for both identification and quantification of protein. AAA involves hydrolyzing the peptide bonds to free individual amino acids, which are then separated, detected, and quantified, using purified amino acids as standards. Nonetheless, UV-visible spectroscopy remains the most widely spread, cost- and time-efficient technique for total protein concentration determination.

To take full advantage of AAA measurement even in the absence of tyrosine and tryptophan residues, the test solution can be to use FTIR-based protein quantification and AAA measurements at first, and then generate concentration calibration curves for the protein of interest in correlation with UV absorbance (at 280 nm or another wavelength). These calibration curves can then be used to determine the concentration of subsequent samples directly by UV absorbance spectroscopy.

UV Spectroscopy Measurement: Detecting Proteins With Absorbance at 205 nm

Determination of protein concentration by measurement of absorbance at 205 nm (A_{205}) is based on absorbance by the peptide bond. This assay can be used to quantitate protein solutions with concentrations of 1–100 μg/mL protein.

The quantitation of proteins by peptide bond absorption at 205 nm is more universally applicable than A_{280}. Furthermore, the absorptivity for a given protein at 205 nm is several-fold greater than that at 280 nm. Thus, lower concentrations of protein can be quantitated with the A_{205} method. Most proteins have extinction coefficients at A_{205} for a 1 mg/mL solution of between 30 and 35. However, an improved estimate can be obtained by taking into account variations in tryptophan and tyrosine content of the protein to be quantitated ($\varepsilon^{1\text{mg/mL}} = 27.0 + 120 \times (A_{280}/A_{205})$). Absorbance at 205 nm can be used to quantitate dilute solutions, or for short path length applications. For example, continuous measurement in column chromatography, or for analysis of peptides where there are few, if any, aromatic amino acids. The disadvantage of this method is that some buffers and other components also absorb at 205 nm (Table 8.2).

Spectrofluorometric Measurement: Detecting Proteins by Intrinsic Fluorescence Emission

Protein concentration can also be determined by measuring the intrinsic fluorescence based on fluorescence emission by the aromatic amino acids tryptophan, tyrosine, and phenylalanine. Usually tryptophan fluorescence is measured as the indication of the fluorescence intensity of the protein sample solution. The concentration of the protein sample solution can be calculated from a calibration curve based on the fluorescence emission of standard solutions prepared from the purified protein. This assay can be used to quantitate protein solutions with concentrations of 5–50 μg/mL. Normally, the excitation wavelength is set to 280 nm and the emission wavelength to between 320 and 350 nm. If the exact emission wavelength is not known, determine it empirically by scanning the standard solution with the excitation wavelength set to 280 nm.

COLORIMETRIC PROTEIN ASSAY TECHNIQUES

All colorimetric protein assays require protein standard to estimate the concentration of a sample. The ideal protein standard to use in a quantitative assay is the exact same protein in a matched matrix/solution that has been assigned using a higher order method, for example AAA or gravimetric analysis. Gravimetric analysis is prone to errors due to the extensive dialysis and drying to remove water and salts from commercial preparations. In practice, there is not always a matched protein standard available. On the other hand, some commercially available standards may be suitable for use. The most commonly used standards are bovine serum albumin (BSA), bovine gamma globulins, and immunoglobulins (used for antibody quantitation). The use of a BSA standard can give misleading results in many assays, especially those methods that are sensitive to the protein sequence, that is, where the signal is generated by specific amino acids. Assays with a low protein sequence dependence will give better estimates when BSA calibration is compared to AAA assignment. This is because AAA assignment quantitates the amount of specific amino acids present following protein hydrolysis and separation; using peptide sequence information, the amount of target protein can then be calculated.

Biuret Reaction

The biuret reaction is based on the complex formation of cupric ions with proteins. In this reaction, copper sulfate is added to a protein solution in strong alkaline solution. A purplish-violet color is produced, resulting from complex formation between the cupric ions and the peptide bond. The biuret reaction with proteins is independent of the composition of the protein; therefore, protein composition is not a factor. However, protein purity and association state could influence the results obtained with the biuret reagent.

The biuret reaction is somewhat insensitive compared with the other methods of colorimetric protein determination. A "reverse biuret method," which is a modification of the biuret reaction, significantly increases sensitivity. In the "classic" biuret reaction, color is produced by the formation of a protein-copper-tartrate complex; in the "reverse" biuret reaction, color is generated by the reduction of excess cupric ions, not bound in the biuret complex, by ascorbic acid to cuprous ions, which are subsequently measured as a complex with bathocuproine. The amount of Cu^{+}-bathocuproine complex is inversely proportional to the protein concentration. The sensitivity of this reaction is greater than either the Lowry assay or the Bradford method. As with the direct biuret reaction, there is no dependence on protein composition. Solution constituents such as Tris buffer, ammonium ions, sucrose, primary amines, glycerol, and dextran can interfere with the biuret reaction.

Lowry Method

The Lowry protein assay is based on the biuret reaction with additional steps and reagents to increase the sensitivity of detection. In the biuret reaction, copper interacts with four nitrogen atoms of peptides to form a cuprous complex. Lowry adds phosphomolybdic/phosphotungstic acid also known as Folin-Ciocalteu reagent. This reagent interacts with the cuprous ions and the side chains of tyrosine, tryptophan, and cysteine to produce a blue-green color that can be detected between 650 nm and 750 nm. The protein detection range is 5–100 μg.

Although the Lowry method uses standards for calibration, which can be a source of error as the composition of the protein of interest may not necessarily match that of the protein standards, it is almost 100-fold more sensitive than determining absorbance at 280 nm. One disadvantage of the Lowry method is that many common substances, such as K^+, Mg^{2+}, NH_4^+, EDTA, Tris-HCl, carbohydrates, and reducing agents, interfere with the method. Furthermore, the Folin reagent is reactive for only a short period of time after addition. The method is complicated and requires more steps and reagents than the BCA or Bradford assays, and this method is destructive to proteins: once the protein sample has reacted with the dye, the protein cannot be used for other assays.

Bradford Method

The Bradford method uses the binding of Coomassie brilliant blue G-250 dye to proteins, which results in a dye-protein complex with increased molar absorbance for the determination of protein concentration. The Coomassie brilliant blue G-250 dye is protonated and is reddish/brown with an absorbance maximum of 465 nm at acidic pH. Under acidic conditions, the dye reacts primarily with arginine and to a lesser extent with lysine, histidine, tyrosine, tryptophan, and phenylalanine residues in proteins, producing a blue color with an absorbance maximum at 595 nm (the absorption range is between 575 nm and 615 nm), and 0.2–20 μg of protein can be detected. The method is the easiest and fastest of the protein determination methods.

The Bradford Coomassie Blue G-250 assay is fast, simple and sensitive. The Coomassie brilliant blue G-250 is stable for long periods of time. Furthermore, the volume of reagents can be reduced and the assay can be performed in a 96-well plate. However, the Bradford Coomassie brilliant blue G-250 dye stains cuvettes, although cuvettes can be cleaned by washing with a dilute SDS solution. Dye binding depends on the basic amino acid content that can vary between proteins. It is, however, useful as a general, sensitive, semiquantitative assay for proteins. With the selection of an appropriate standard, the assay can be both accurate and sensitive. It has been demonstrated that decreasing the acidity in the reaction by the addition of NaOH, the inclusion of Triton X-100, or the addition of SDS can decrease protein-to-protein variability.

Another issue for this method is that the concentrated protein solutions can form a precipitate upon contact with the dye reagent. If this is observed, the protein solution should be diluted to determine protein concentration. Furthermore, this method is also destructive to proteins, meaning that once the protein sample has reacted with the dye, the protein cannot be used for other assays.

Bicinchoninic Acid Assay (BCA) Method

The BCA assay method is based on the fact that the sodium salt of bicinchoninic acid reacts with the cuprous ion generated by the biuret reaction under alkaline conditions. The bicinchoninic acid cuprous complex forms a deep blue color that is read at 562 nm, and the detection range is 0.2–50 μg. The BCA reagent is stable under alkaline conditions, so it can be included in the biuret alkaline copper solution. The development of color in the BCA assay depends on time, temperature, and pH. The assay can be performed at room (ambient) temperature, but increases the reaction temperature can significantly reduces the time required for maximal color development and increases sensitivity. The assay is compatible with detergents, and therefore is better than both the Lowry and Coomassie dye assays.

BCA method has less protein/protein variability than the Bradford assay. The variation as a function of protein composition could be decreased by reaction at 60°C. Furthermore, the volume of reagents can be reduced, and it can be performed in 96-well plates. However, reducing reagents that reduce the cupric ions to cuprous ions interfere with the assay, and the chelating agents (e.g., EDTA) that chelate copper interfere as well. H_2O_2 can also interfere with the BCA assay. Phospholipid interferes with the BCA protein assay, resulting in artificially high values that reflect the interaction of phospholipid with the BCA reagent to yield a chromophore absorbing close to 562 nm. Like the Bradford method, this method is destructive to proteins. Therefore, once the protein sample has reacted with the copper, the protein cannot be used for other assays.

Standards and Assay Validation

Any assay for protein concentration must be validated to ensure an accurate value for the sample of interest. The selection of an appropriate standard is of critical importance because, with the possible exception of the biuret reaction, the protein assays described earlier all depend on the quality of the protein for the response. The selection of an appropriate standard together with a rigorous validation of the analytical process can solve the problems presented by the protein composition. The validation methodology must include measurements to ensure that the sample protein concentration is within the dynamic range of the assay. Furthermore, all sample types must be independently validated to assess the effect of protein composition as well as interfering or enhancing substances that may be contained in the sample.

FLUORESCENT DYE-BASED ASSAYS

Microplate Detection Method

Amine-labeling "derivatization" using various fluorescent probes is a common technique to quantitate amino acid mixtures in amino acid analysis. This technique can be used to quantitate proteins and peptides containing either lysine or a free N terminus, both of which need to be accessible to the dye. Upon reaction with amines, the dyes display a large increase in fluorescence that, for part of the dynamic range, will generate a linear response with increasing protein concentration. The range of this method is 0.05–25 μg. Three dyes used to quantitate proteins, or amino acids in a microplate format include *o*-phthalaldehyde (OPA), fluorescamine, and 3-(4 carboxybenzoyl)quinoline-2-carboxyaldehyde (CBQCA). Fluorescamine reacts directly with the amine functional group, whereas OPA and CBQCA require the addition of a thiol (2-mercaptoethanol) or cyanide.

Cuvette Detection Method

In a cuvette-based format, OPA reacts with primary amino acid (except cysteine) in the presence of 2-mercaptoethanol or 3-mercaptopropionic acid to form a highly fluorescent adduct. OPA also reacts with 4-amino-1-butanol and 4-aminobutane-1,3-diol produced from oxidation of proline and 4-hydroxyproline, respectively, in the presence of chloramine-T plus sodium borohydride at 60°C, or with *S*-carboxymethyl-cysteine formed from cysteine and iodoacetic acid at 25°C. Fluorescence of OPA derivatives is monitored at excitation and emission wavelengths of 340 nm and 455 nm, respectively. Detection limits are 50 fmol for amino acid.

The Limitations of Using Fluorescent Dyes

All three dyes offer improved sensitivity and dynamic range when compared with absorbance-based protein quantitation assays. OPA is generally preferred over fluorescamine because of its enhanced solubility and stability in aqueous buffers. The use of amine-derivatization agents for protein quantitation is limited, as the assay displays a large protein-to-protein variability due to variation in the number of lysine residues in proteins, requiring the need for a "matched" standard. Assay interference from glycine and amine containing buffers, ammonium ions, and thiols common in many biological-buffering systems limits the application of such assays. The reproducibility of the assay is dependent on the pH of the reaction. For example, protein samples that contain residual acids could reduce the rate of amine derivatization.

THE CHOICE OF MEASUREMENT FORMAT

Cuvette Measurement Method

The format used in the measurement depends on the throughput, sensitivity, and precision required of the assay. Traditionally, cuvettes have been used for the majority of spectrophotometric protein assays. Quartz cuvettes can be costly, therefore, glass cuvettes are preferred. However, both of these may have to be washed between measurements to remove dye and adsorbed protein. Disposable plastic cuvettes are available and can be used to increase the throughput where many samples have to be measured, or where the reagent is prone to sticking to the cuvette surface like in Bradford reagent. Staggering of sample analysis is especially important if the signal is not stable or does not run to completion within the time frame of the assay like in BCA or Lowry assays. Replacing the cuvette in the holder between each measurement because of cleaning, or the use of disposable cuvettes, can result in changes in alignment, resulting in significant changes in the amount of light reaching the detector. This is especially important if low-volume cuvettes are being used, where the transmission window is reduced in size.

Cuvette Cleaning Method

Care should be taken when handling and cleaning cuvettes. It is also important to prevent fingerprints from contaminating the transmitting surfaces. Cuvettes should be washed with either water or an appropriate solvent between runs and dried using a stream of nitrogen gas. If smearing of the transmitting surface is observed, the cuvette can be rewashed in water, ethanol, and finally acetone, or smears can be removed using ethanol and lint-less lens tissue. If protein deposition is a recurring problem, cuvettes can be soaked overnight in nitric acid and thoroughly washed before use.

Microplate Method

The use of microplate for protein assay can enhance speed and throughput and decrease sample and reagent usage. Many of the commercial fluorescent assays are specifically designed for plate formats. The plate reader format also offers the advantage of being able to read multiple samples within a short period (typically 25 s), reducing potential timing differences in reactions that do not go to completion or are unstable. Quartz 96-well plates tend to be expensive, difficult to clean, and prone to scratches that can affect light transmission. Since the cost decreases, disposable plastic plate can be the alternative.

Preparation of Samples for Measurement

Care should be taken in the preparation of protein assays in plate formats. The use of lower volume samples (down to 5 μL for some assays) can increase the relative pipetting errors of high-viscosity solutions. Well-to-well contamination should be avoided by using fresh pipette tips for each sample and reagent. Regular calibration of the instrument should be performed using either optical standards or solid-phase fluorescent standard plates to ensure equal transmission/light detection from all wells.

MOLECULAR SEPARATION OF PROTEIN MOLECULES THROUGH COLUMN CHROMATOGRAPHY

After the cells are broken and the cell extracts are released, proteins can then be further purified through various biochemical methods. The most common method for purifying proteins from other protein molecules within a given sample is column chromatography. Basically, this method uses a glass or plastic tube filled with resin that can separate proteins based on their physical properties as they flow through the column. At first, the protein sample is applied to the top of the column. As a buffer solution is flowed continuously through the column, proteins in the sample migrate through the column at different rate, depending on the nature of the matrix and the physical and the chemical properties of the proteins. Therefore, the targeted protein can be separated from others.

Size Exclusion Chromatography

This chromatography can separate proteins based on the size and shape on a gel filtration column (Fig. 8.3). The column matrix also known as resin consists of microscopic beads of inert material. The resin bead has many tiny pores like a whiffle ball (Fig. 8.4). In this type of column, small molecules, which is smaller than the cut-off threshold size, are more likely to go through the pore of the matrix, and thus are trapped in the resin and travel through the column more slowly. Larger molecules, on the other hand, move through the column more quickly because they cannot fit into the beads and are excluded from entering the pores in the beads. These larger molecules can only pass through the spaces between resin beads, so they travel a shorter distance overall. Thus, large molecules will emerge first from the column, and small molecules will emerge last.

Ion-Exchange Chromatography

Ion-exchange chromatography uses a resin to separate proteins according to their surface charges. This type of column contains a resin bearing either positively or negatively charged chemical groups. Resins containing positively charged groups attract negatively charged solutes and are referred to as anion-exchange resins (Table 8.3). Resins with negatively charged groups are cation exchangers. In low-salt solutions, proteins with a negative surface charge will bind more strongly

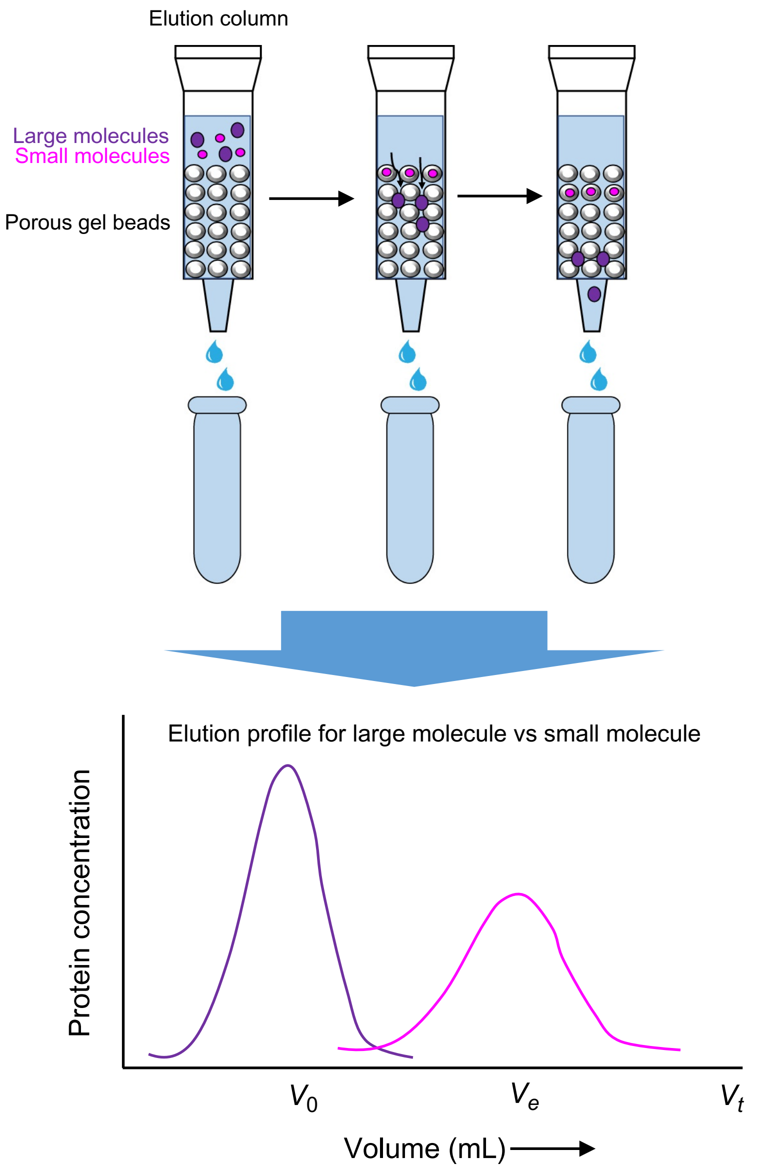


FIG. 8.3 Size exclusion chromatography in which small proteins (under the cutoff limit) enter pores of beads and take a longer route to exit the column than large proteins, which are excluded from porous beads and quickly exit column first.

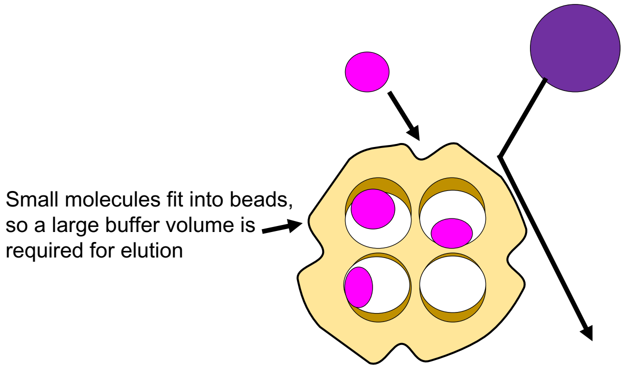
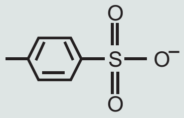
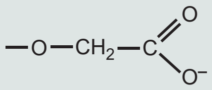
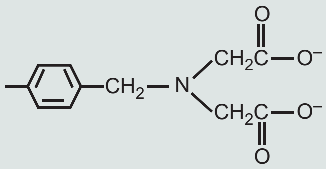
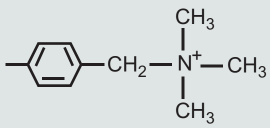
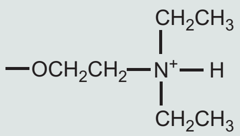


FIG. 8.4 Size exclusion chromatography column. Larger molecules are excluded from gel beads and emerge from the column sooner while smaller molecules must navigate the intricate network of pores and exclude later.

TABLE 8.3 Cation- and Anion-Exchange Resins Commonly Used for Biochemical Separations

Cation-Exchange Media	Structure
Strongly acidic, polystyrene resin	
Weakly acidic, carboxymethyl (CM) cellulose	
Weakly acidic, chelating, polystyrene resin (Chelex-100)	
Anion-Exchange Media	**Structure**
Strongly basic, polystyrene resin (Dowex-1)	
Weakly basic, diethylaminoethyl (DEAE) cellulose	

to positively charged anion-exchange columns (Fig. 8.5). Likewise, proteins with a positive surface charge will bind to negatively charged cation-exchange columns. Later, higher salt concentration buffer is applied to the column so it can compete with the resin for the bound proteins, and the bound proteins can then be eluted through the high-salt buffer.

Affinity Chromatography

Affinity chromatography uses the principle that the protein binds to a molecule for which it has specific affinity. This is because in most instances proteins carry out their biological activity through binding or complex formation with specific small molecules, or ligands. This small molecule can be immobilized through covalent attachment to the resin in a column (Fig. 8.6). When the sample goes through the column, the protein of interest, in displaying affinity for its ligand, become bound and immobilized itself. The protein of interest is thus removed from the mixture of sample. Finally, the protein is dissociated or eluted from the resin by the addition of high concentrations of free ligand in solution. Because this method relies on the biological specificity of the protein of interest, it is a very efficient procedure. A typical example is the resins coupled to an antibody that recognizes a specific protein, or it may contain an unreactive analogue of an enzyme's substrate. The power of affinity chromatography lies in the specificity of binding between the affinity reagent on the resin and the molecule to be purified. As such, it is possible to design an affinity chromatography procedure to purify a protein in a single step.

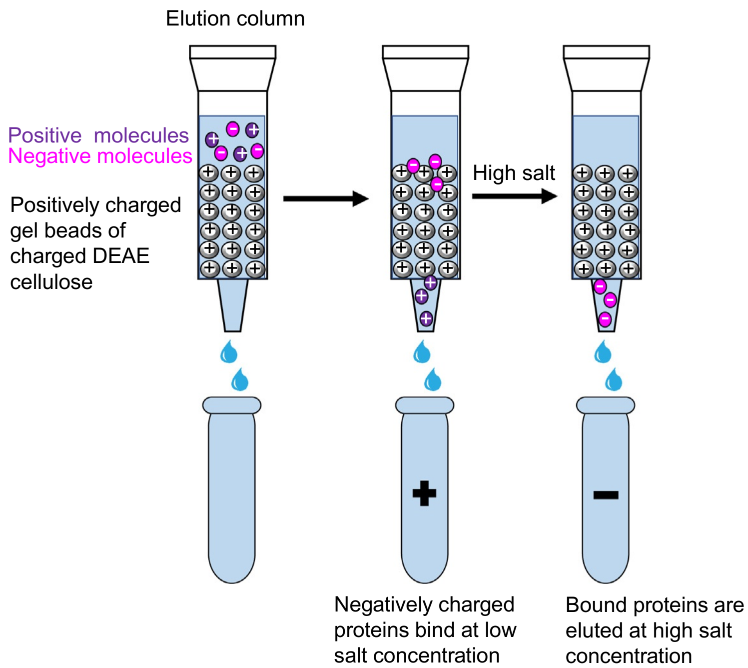


FIG. 8.5 Ion exchange chromatography separates molecules based on positive and negative charges. If the beads in column are positively charged, then negatively charged proteins will bind to column matrix at low salt as a result of ionic interactions. Proteins can then be induced to dissociate with high salt.

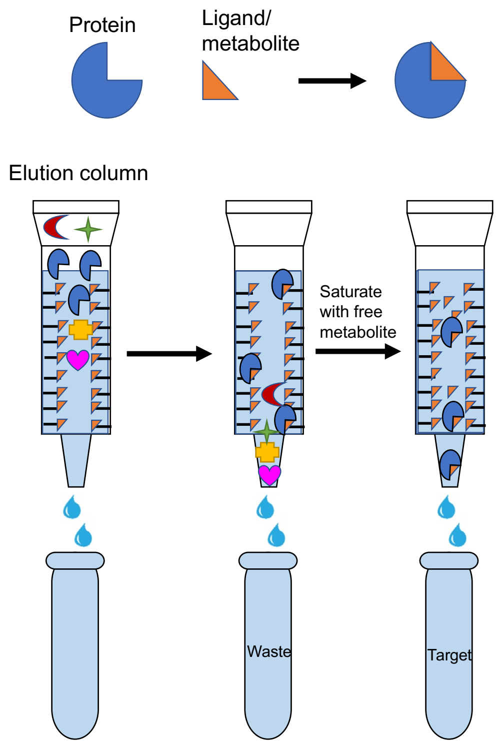


FIG. 8.6 Affinity chromatography to isolate molecules based upon ligand binding abilities.

Hydrophobic Interaction Chromatography

In this purification technique, proteins passed through a chromatographic column packed with a support resin to which the hydrophobic groups are covalently linked. Phenyl Sepharose, which contains a phenyl group, is commonly used in the column. Usually, the protein sample is prepared in high-salt buffer. When the protein sample goes through the column, proteins bind to the phenyl group by virtue of hydrophobic interactions. Proteins in a mixture can be differentially eluted from the phenyl group by using low salt concentration buffer or by adding solvents such as polyethylene glycol to the elution fluid.

MOLECULAR SEPARATION THROUGH ELECTROPHORESIS

Sodium Dodecyl Sulfate-Polyacrylamide Gel Electrophoresis (SDS-PAGE)

Prior to any downstream analysis, purity and integrity are the very first qualities that need to be assessed for any protein sample. This is routinely achieved by SDS-PAGE. This technique, associated with staining method, can detect bands of protein in a simple and relatively rapid manner (just a few hours).

The hydrophobic tail of SDS interacts strongly with polypeptide chains. The number of SDS molecules bound by a polypeptide is proportional to the length (the number of amino acid residues) of the polypeptide. Each sodium dodecyl sulfate contributes two negative charges (Fig. 8.7). Collectively, these charges overwhelm any intrinsic charges that the protein might have. Therefore, SDS has two advantages. (1) It coats all the polypeptides with negative charges, so they all electrophorese toward the anode. (2) It masks the natural charges of the subunits, so they electrophorese according to their molecular masses and not by their native charges. Small polypeptides fit easily through the pores in the gel, so they migrate rapidly. Large polypeptides migrate more slowly. SDS is also a detergent that disrupts protein tertiary structure. Both heat and reducing agent such as DTT and 2-mercaptoethanol can also denature the protein and break the covalent bonds between subunits.

After reduction and denaturation by SDS, proteins migrate in the gel according to their molecular mass, allowing to detect potential contaminants, proteolysis events, and so forth (Fig. 8.8A). The electrophoretic mobility of proteins upon SDS-PAGE is inversely proportional to the logarithm of the protein's molecular weight (Fig. 8.8B). SDS-PAGE is often used to determine the molecular weight of proteins.

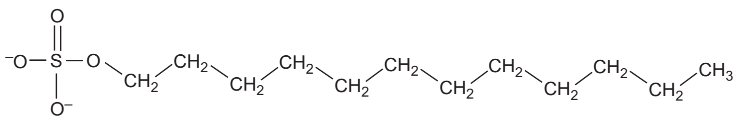


FIG. 8.7 Chemical structure of sodium dodecylsulfate (SDS).

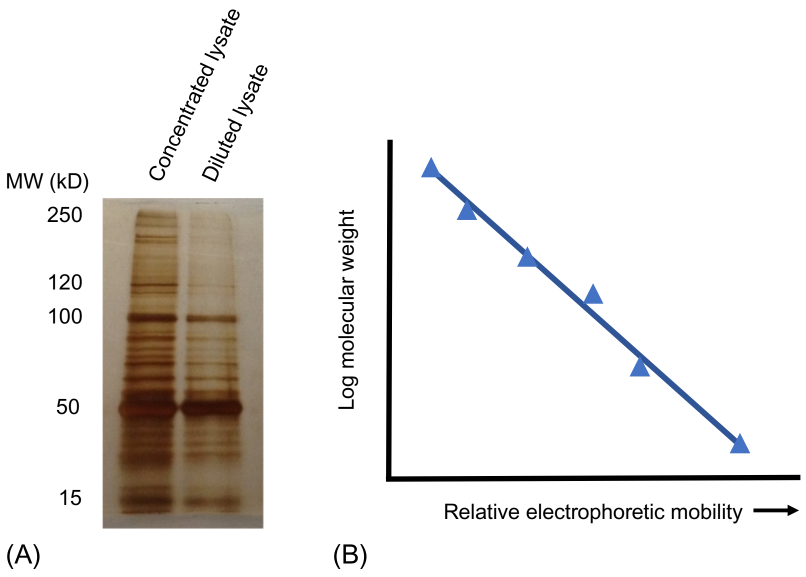


FIG. 8.8 (A) An example of SDS-polyacrylamide gel electrophoresis coupled with a silver stain to visualize polypeptides. (B) Relative SDS-PAGE electrophoretic mobility is inverse relative to the log of the molecular weights of the individual polypeptides.

Gel Staining After Electrophoresis: Colorimetric Staining Methods

Three high-sensitivity colorimetric staining methods can be used either directly after electrophoresis, including Coomassie blue staining, zinc-reverse staining, and silver staining. These can detect as low as 10-ng and 1-ng protein bands, respectively.

Staining with the organic dye Coomassie Brilliant Blue is the most frequently employed method for protein detection in SDS-PAGE gels. This anionic triphenylmethane dye is used in two modifications: Coomassie R-250 (Red tint) and Coomassie G-250 (Green tint), which has two additional methyl groups. In the presence of an acidic medium, these dyes stick to the amino groups of the proteins by electrostatic and hydrophobic interactions. The advantages of Coomassie staining methods are quantitative binding of the dye to proteins, low price, and good reproducibility. For mass spectrometric analysis, the dye can be removed from the gel with bicarbonate prior to tryptic digestion. When a protein is detectable with Coomassie Brilliant Blue, enough protein is present for appropriate mass spectrometry (MS) analysis. The disadvantages are the long staining times, the relatively low sensitivity, and the narrow dynamic range.

Zinc-reverse staining, also known as negative staining, uses imidazole and zinc salts for protein detection in electrophoresis gels. It is based on the precipitation of zinc imidazole in the gel, except in the zones where proteins are located. When zinc reverse staining is applied on a Coomassie Blue-stained gel, previously undetected bands can be spotted. This technique is rapid, simple, cheap, and reproducible and is compatible with MS analysis.

Silver staining is based on the binding of silver ions to the proteins followed by reduction to free silver, sensitization, and enhancement. If silver staining is used as a second staining, it is essential to fix the proteins in the gel with acidic alcohol prior to initial Coomassie Blue staining. Detecting proteins with silver staining is widely used, because the sensitivity is below 1 ng per spot, and the costs for reagents are relatively low. Although it is a multistep procedure, the results are available relatively quickly. Ideally, the spots are dark brown to black on a light beige background (Fig. 8.9). Two drawbacks of this technique are that proteins are differentially sensitive to silver staining and that the process may irreversibly modify them, preventing further analysis. In particular glutaraldehyde, which is generally used during the sensitization step, may interfere with protein analysis by MS due to the introduction of covalent cross-links. To circumvent this problem, a glutaraldehyde-free modified silver-staining protocol has been developed, which is compatible with both matrix-assisted laser desorption/ionization (MALDI) and electrospray ionization-MS.

Gel Staining After Electrophoresis: Fluorescence Staining Method

Fluorescence dyes give very wide linear dynamic ranges, over four orders of magnitude. Since fluorescent staining is an endpoint method, the results are highly reproducible. Several fluorescent dyes such as Nile Red, ruthenium(II) tris(bathophenantroline disulfonate) (RuBPS), SyPro, and Epicocconone, can also be used to reveal a few nanograms of proteins in gels. Although CyDyes can even reveal as low as 1 ng of protein, it is inconvenience because it has to be incorporated before gel electrophoresis. On the other hand, the sensitivities of SyPro Orange, Red, and Tangerine are similar to that of Coomassie Blue, SyPro Ruby can detect down to about 1 ng of protein in a spot. Deep Purple contains the fluorophore "epicocconone" from the fungus *Epicoccum nigrum* and is even more sensitive than SyPro Ruby, down to a few hundred picograms. Besides the high sensitivity, it does not create speckles in the background and is a natural product, which is easy to dispose of. Apart from Nile Red, these staining methods are compatible with subsequent MS analysis. However, their major disadvantage is that they require a fluorescence imager for visualization and that they are significantly more expensive than classical colorimetric dyes.

GEL-BASED PROTEOMICS: PROTEIN SEPARATION

SDS-PAGE gives very good resolution of polypeptides, but sometimes a mixture of polypeptides is so complex that we need an even better method to resolve them all. The field that uses techniques to resolve thousands of polypeptides is called proteomics. Proteomics tools have been exploited to address many molecular biological questions related to various healthy and disease states. Both gel-based proteomics, including two-dimensional gel electrophoresis (2DE) gel and 2DE fluorescence differential imaging gel electrophoresis (DIGE), and gel-free proteomics, such as liquid chromatography (LC) and capillary electrophoresis (CE), are useful methods to identify proteins. Currently, proteomics has been dominated by 2DE gel coupled with MS, as it is still the most reproducible and effective technology to separate overall proteins of microorganisms, cells, and heterogeneous tissues.

Two-Dimensional Gel Electrophoresis

Gel-based proteomics is the most popular and well-established technique for global protein separation and quantification. Through this technique, overall protein expression of tissues can be analyzed on a large scale, and it is a cheaper approach than gel-free proteomics. 2DE gel, MS, and bioinformatics tools are the key components of gel-based proteomics. The most important steps involved in the 2DE gel technique are summarized in Fig. 8.9. In 2DE, the complex protein samples are separated in two dimensions according to their net charge at different pH and their molecular weights determined by SDS-PAGE. Protein migration in a perpendicular direction provides a spot map of the proteins distributed in 2DE gel. The technique has great resolving power, making it possible to visualize >10,000 spots corresponding to >1000 proteins. After the separation, the next typical steps of 2DE gel technique are spot visualization, spot evaluation, and expression analysis. The final step is protein identification by MS.

First Dimension Electrophoresis: Conventional Isoelectric Focusing

In 2DE, the complex protein samples are separated in two dimensions according to their net charge at different pH, which is each protein's isoelectric focusing (IEF), and their molecular weights through SDS-PAGE. In the first step, the mixture of proteins is electrophoresed through a narrow tube gel containing molecules called ampholytes that set up a pH gradient

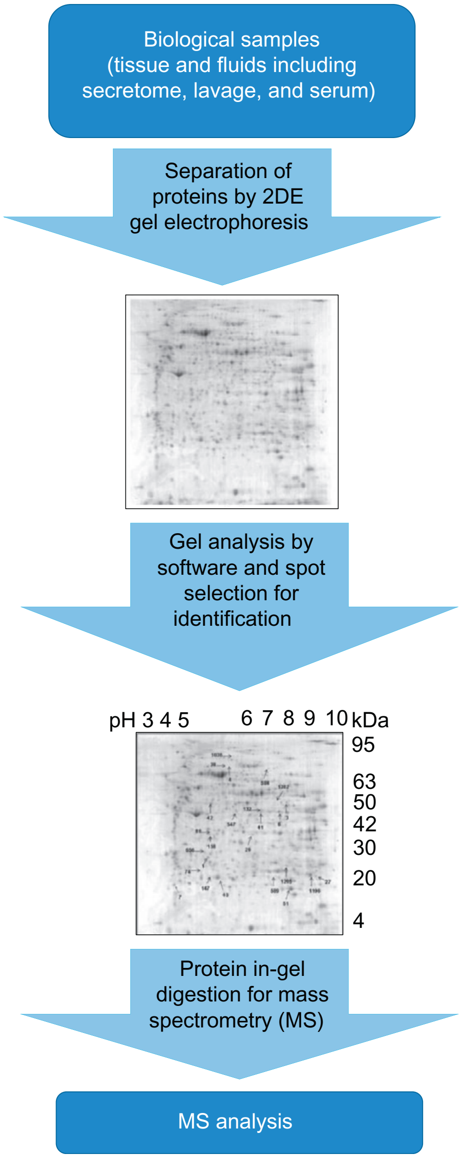


FIG. 8.9 Application of 2DE gel electrophoresis for mass spectrometry analyses.

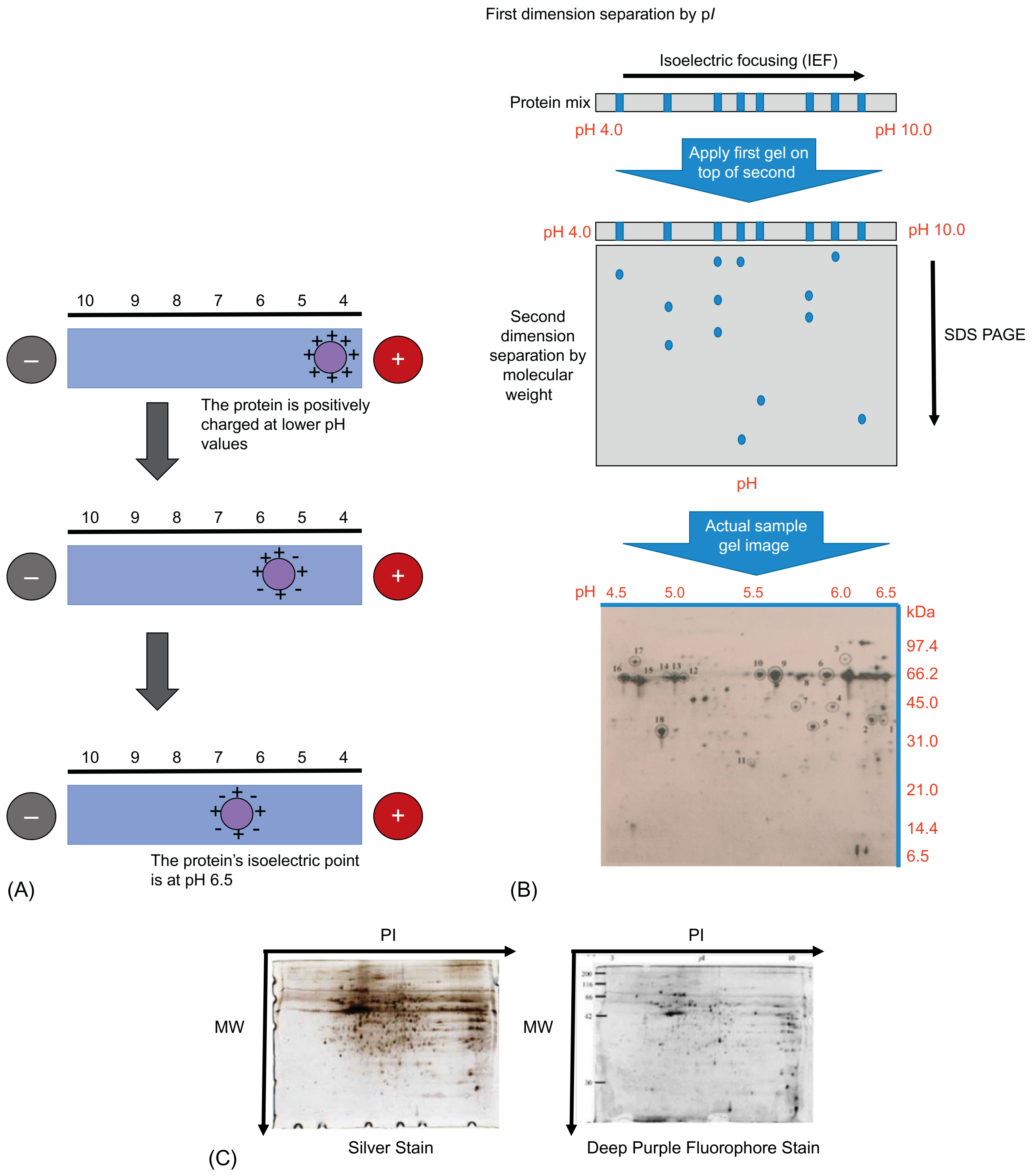


FIG. 8.10 (A) An isoelectric focusing gel with a pH gradient. Proteins migrate through the gel until they reach a pH that matched their p*I*, at which point they are no longer charged and therefore stop migration.(B) Two-dimensional gel electrophoresis separation of proteins by p*I* and molecular weight. (C) An example of two dimensional polyacrylamide gel electrophoresis of mouse colon protein stained by silver staining or deep purple flurophore dye.

from one end of the tube to the other. A negatively charged protein will electrophorese toward the anode. This is because the protein has an overall negative charge after treatment with heat, SDS, and reducing agents. As it migrates through a gradient of increasing pH, however, the protein's overall charge will decrease until the protein reaches the pH region that corresponds to its isoelectric point (p*I*) (Fig. 8.10). Since the p*I* is the pH at which the protein has no net charge, without

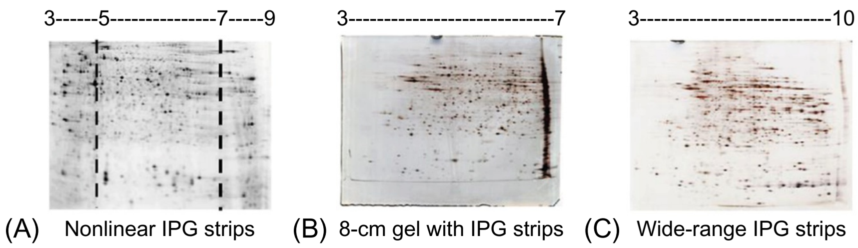


FIG. 8.11 Various immobilized pH gradient strips used in the first dimension step of 2-DE gels. (A) Gel with nonliner (NL) IPG strips (p*I* 5–7). (B) 8 cm gel with IPG strips (p*I* 3–7). (C) Wide-ranged (p*I* 3–10) IPG strip.

net charge, protein is no longer drawn toward the anode or the cathode, and thus it stops. IEF focuses proteins at their isoelectric point in the gel. Although this conventional method is easy to prepare and do not require much casting equipment, it has the disadvantage that the ampholytes have some susceptibility to flow toward the cathode, and this gradient flow usually causes a reduction in reproducibility.

First Dimension Electrophoresis: Immobilized pH Gradient (IPG)

An immobilized pH gradient strip (IPG) is an integrated part of a polyacrylamide gel matrix fixed on a plastic strip. Copolymerization of a set of nonamphoteric buffers with different chemical properties is included. Ready-made IPG strips are available with different lengths and p*I*. Usually, short IPG strips are used for fast screening while longer ones are used for maximal and comprehensive analysis. Various IPG gels are shown in Fig. 8.11. A commercial precast acrylamide gel matrix copolymerized with a pH gradient on a plastic strip results in a more stable pH value than the traditional ampholyte method. The IPG gel has the ability to avoid cationic accumulation and to produce a better-focused protein with less smearing, and it allows the protein molecules to move at different rates across the gel based on their charge and the volt hours setting, which determines speed pattern and reproducibility. The advantages of using IPG strips over ampholytes include reduced cathodic drift, higher mechanical strength as the strips are casted on a plastic backing that minimizes gel breakage, and higher protein loading capacity due to the sample loading method.

First Dimension Electrophoresis: Nonequilibrium pH Gel Electrophoresis

Nonequilibrium pH gel electrophoresis (NEPHGE) technique can resolve proteins with basic to extremely high p*I* (7.0–11.0) that cannot be separated by traditional methods. Compared to NEPHGE techniques, protein loss is higher in the IPG-based method, especially for basic proteins. The reproducibility of spots is slightly better in the NEPHGE-based method than in the IPG method. In general, about half of detected basic protein spots are not reproducible by IPG-based 2DE, whereas the NEPHGE-based method has excellent reproducibility in the basic gel zone. The reproducibility of acidic proteins is similar in both methods.

Second Dimension SDS-PAGE

The second step of 2DE separates proteins based on their molecular weight using a vertical electrophoretic device with either Laemmli buffer or Tris-Tricine buffer. Instead of loading protein sample within the wells, the first-dimension rehydrated strip is carefully placed on the top of the SDS-PAGE and sealed with agarose (Fig. 8.12). Here, the proteins are resolved according to their sizes by SDS-PAGE.

Postelectrophoresis

After 2DE, the proteins are detected by SDS-PAGE with small pore sizes, and the proteins are unfolded. Under this condition, it is easy to prevent the proteins from diffusing into and out of the matrix. Fixing with highly concentrated trichloroacetic acid is therefore not necessary. The gel can be stained as with the standard SDS-PAGE, as described in previous section.

In 2DE, proteins can be prelabeled with a fluorescent dye after the IEF step prior to SDS-PAGE or prior to IEF with monobromobimane. The spots can be directly scanned in the gel (while it is still in the cassette) when low-fluorescent glass is used. The fluorescent label exhibits high sensitivity and wide dynamic range of the signal.

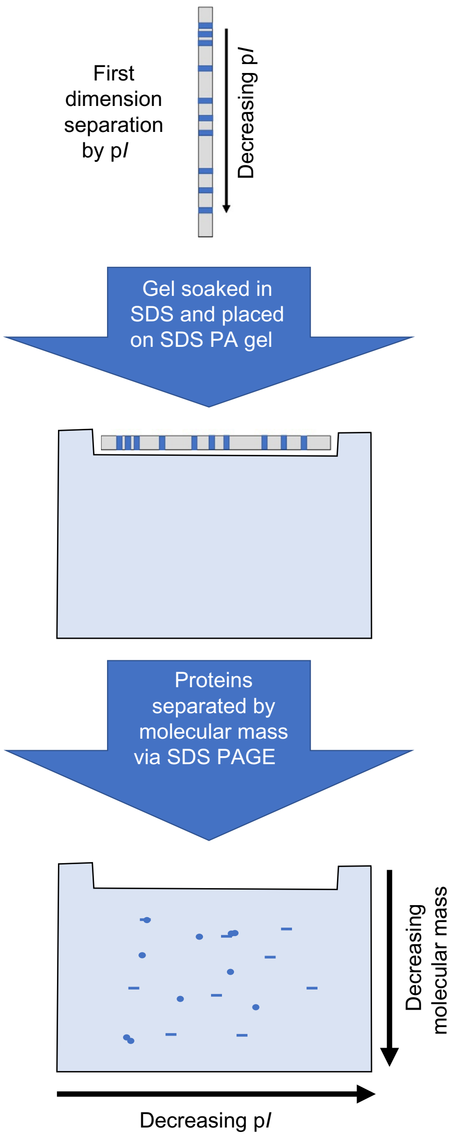


FIG. 8.12 The schematic process of the second step of 2DE which separates proteins based on their molecular weight.

The Analysis of 2DE Images

Analysis of a 2DE gel includes spot visualization, spot evaluation, and expression analysis through bioinformatics software. The key steps of the workflow of most of the automated methods are image quality control, image alignment, spot detection, automatic analysis, editing of spot detection, review the results, statistical analysis, calibration of spots against either a molecular weight ladder or known proteins, spot picking, and importing the protein ID. These automated methods are helpful for gel identification, particularly in quantitative proteomics (Fig. 8.13). The analysis identifies biomarkers by quantifying individual proteins and showing the separation between one or more protein spots on a scanned image of a 2DE gel. Additionally, spots can be matched between gels of similar samples—for example, proteomic differences between early and advanced stages of an illness. Finally, the last step of the analysis is protein identification by MS.

Modification of 2DE: Two-Dimensional Difference Gel Electrophoresis

The development of image technology has introduced the differential imaging gel electrophoresis (DIGE) technique. A quantum leap in 2D gel electrophoresis methodology uses size- and charge-matched cyanine dyes (CyDye, DIGE,

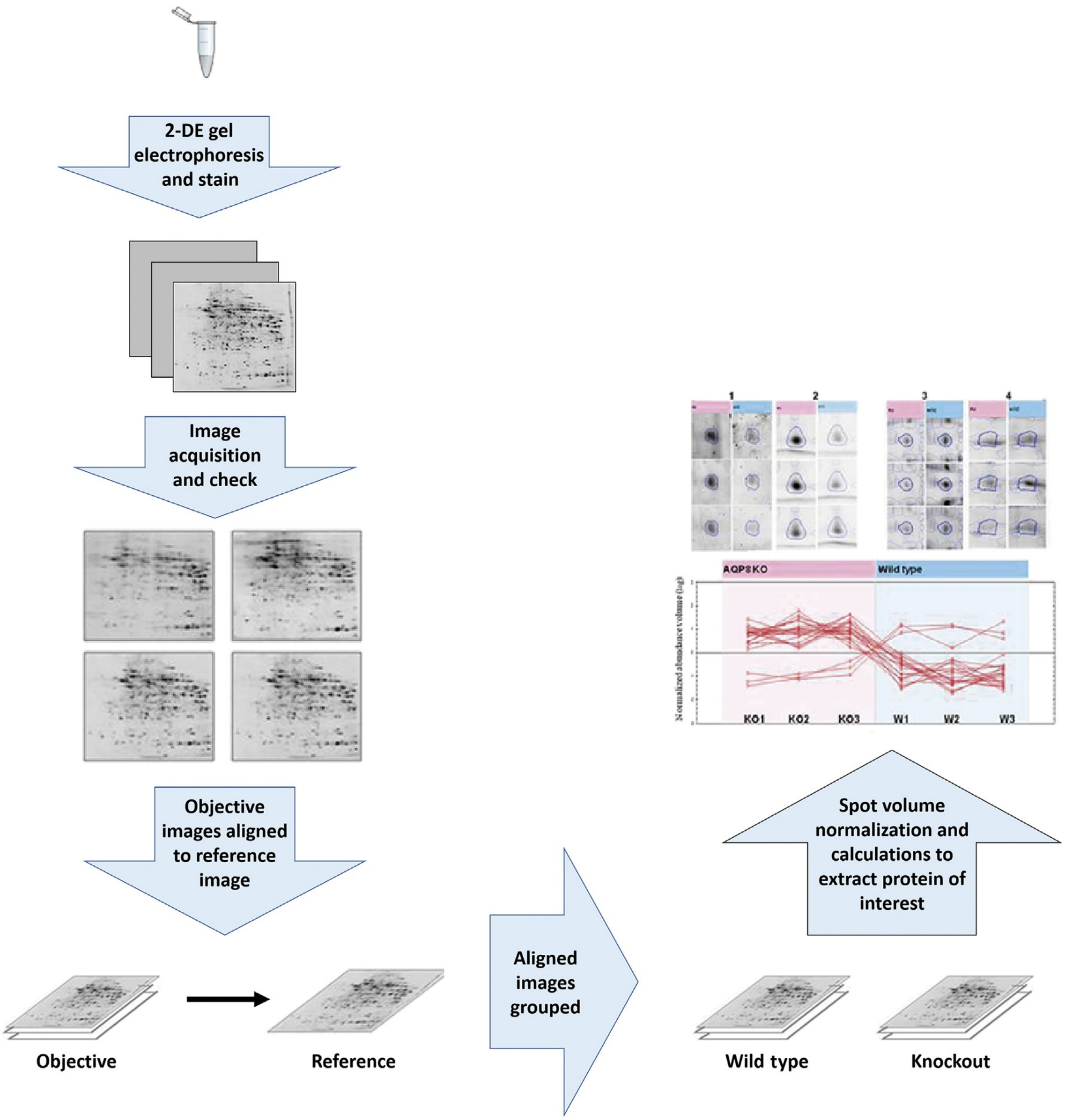


FIG. 8.13 Progenesis SameSpots (nonlinear dynamic) image analysis of a 2-DE gel electrophoresis.

Fluors) with different excitation and emission wavelengths as protein labels for different samples. This method can increase sensitivity and reproducibility of 2DE using multiplexed fluorescent dyes (e.g., Cy2, Cy3, and Cy5) to covalently label protein samples. 2D-DIGE can run more than one sample (maximum 3 samples) on a single gel at once to address the issue of gel-to-gel variability.

In the DIGE technique, different fluorescent cyanine (Cy) dyes are used for labeling proteins from different samples. After mixing these samples in equal ratio and running them together as one sample, the same protein from different samples migrates to the same position on the 2D gel, where it can be easily examined and differentiated by the different fluorophore-labeled dyes and imaged to calculate its abundance. The workflow of 2D-DIGE is shown in Fig. 8.14. Two important factors in 2D-DIGE can affect the sensitivity of the result: minimum labeling by attaching the dye to the free lysine residues, and saturation labeling of all cysteine residues. For the minimum labeling to the lysine residues, only 3%–5% of the total proteins will receive a label, which ensures that only singly labeled proteins will be detected. In this case the labeled proteins comigrate with the nonlabeled proteins in both separation directions, and the resulting 2D image looks like the one

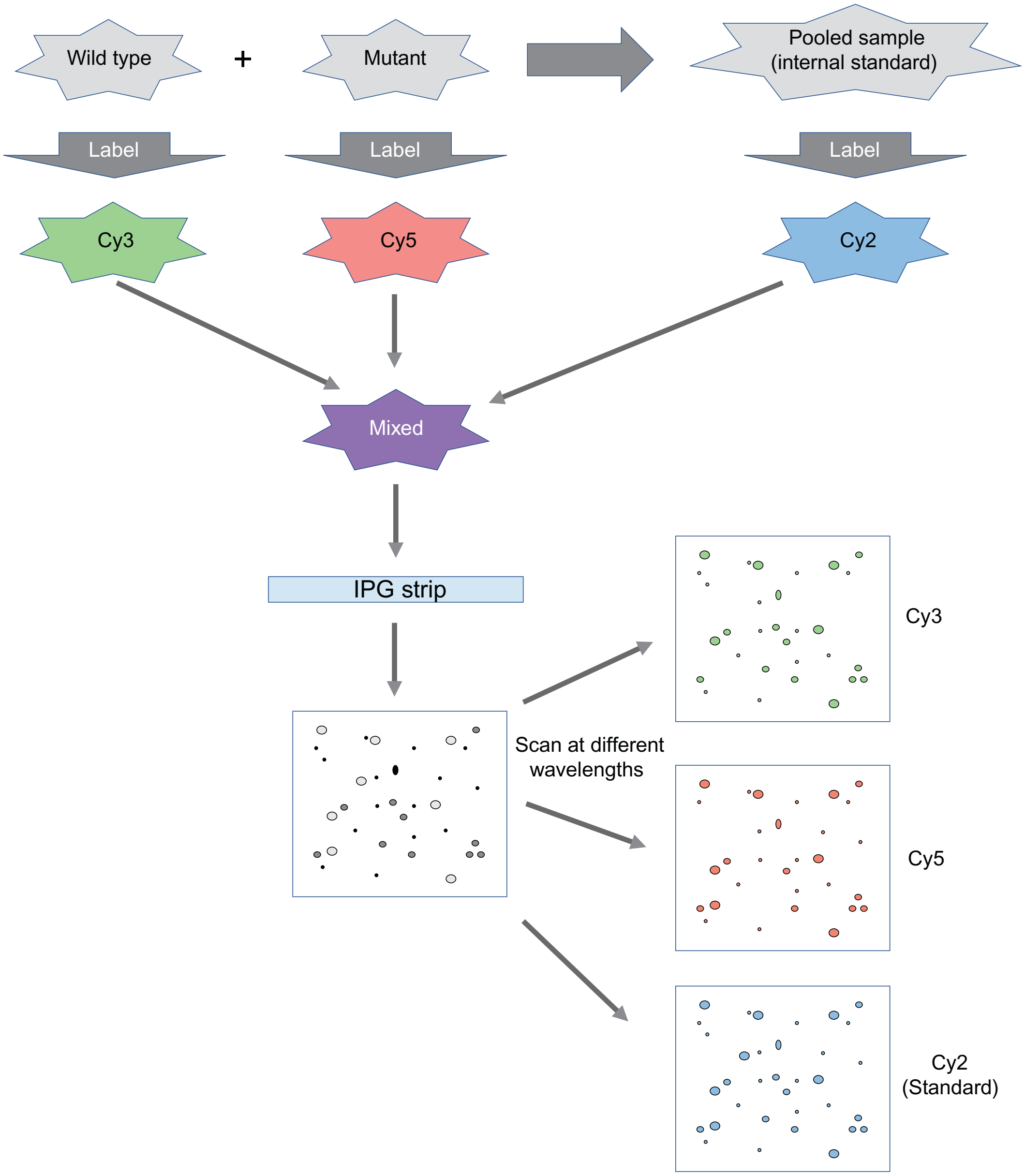


FIG. 8.14 Schematic of 2-D difference gel electrophoresis of two samples in one gel. When more samples are to be compared, more gels are run, every time including the internal standard, which is a mixture of pooled aliquots of all samples.

achieved with poststaining. Typically, 400 pmol/l of one of the three available cyanine dyes is added to 50 μg protein of each sample and the pooled standard. The sensitivity of detection is comparable to a sensitive silver staining method. Meanwhile, a considerable increase in sensitivity is obtained through saturation labeling of cysteine. 2D patterns resulting from as little as 1 μg total protein load can be visualized. Because all available cysteine residues are labeled, many multiply labeled proteins exist. The resulting spot patterns are different from those achieved with poststained or lysine labeled proteins. After scanning the gel at different wavelengths, the comigrated protein spots of different sample origins are codetected using dedicated software, which allows very fast and completely automatic pattern evaluation, thus avoiding any bias introduced by an operator. With the help of the pattern created by the internal standard, which is run in each gel, the patterns of different gels can easily be matched and the sample spot volumes can be normalized to the spot volume of the standard. This results in very accurate protein difference ratios.

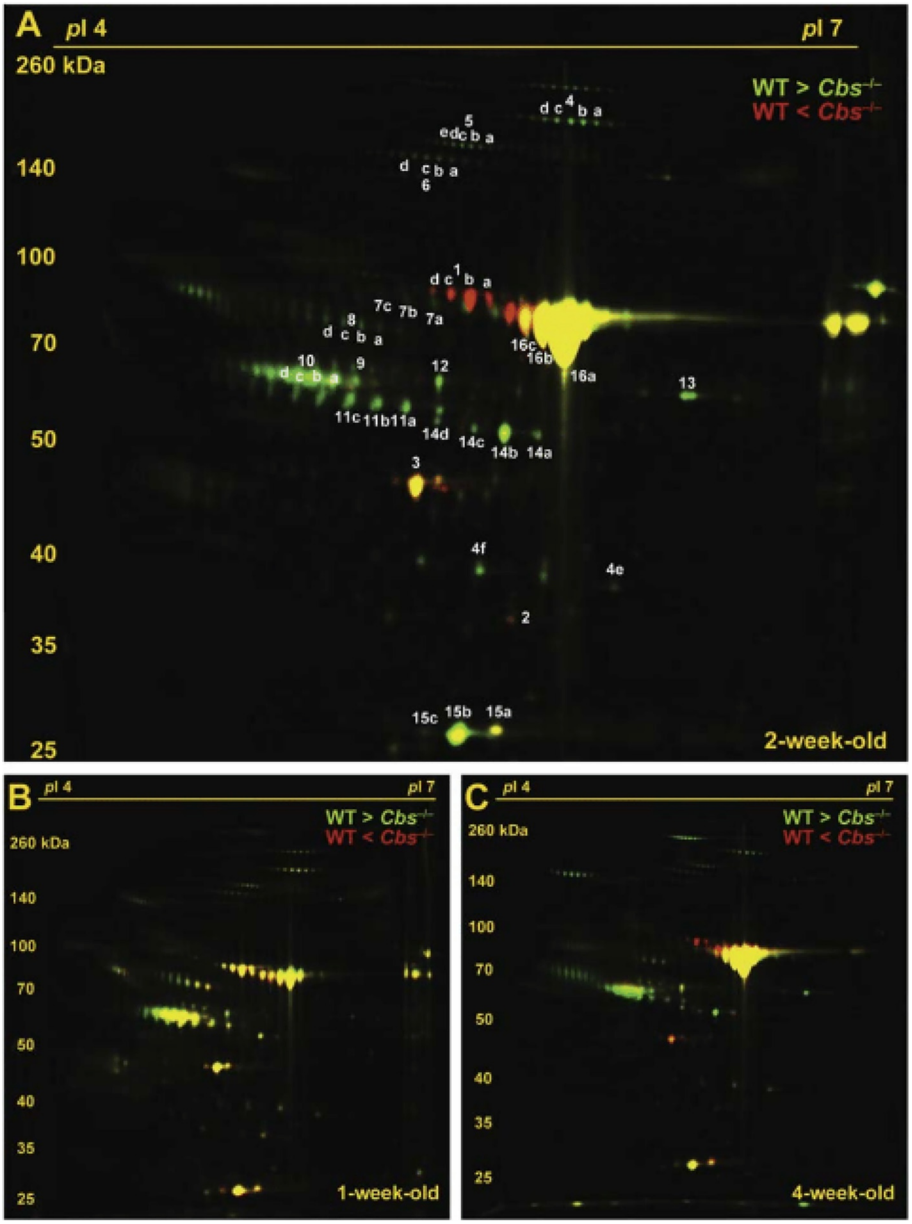


FIG. 8.15 An example of 2D DIGE proteomic analysis of altered plasma protein profiles in juvenile mice.

2D-DIGE is an important tool, especially for clinical laboratories involved in the determination of protein expression levels and disease biomarker discovery. Application of 2D-DIGE can reduce protein ratio errors because of low gel-to-gel variations, which is frequently occurred in the 2DE gel technique. When absolute biological variation between samples is the main objective, as in biomarker discovery, 2D-DIGE is one of the methods of choice. For example, Fig. 8.15 presents 2D-DIGE separation of plasma protein from wild-type mice and cystathionine β-synthase-deficient (Cbs −/−) mice, which is an animal model for homocystinuria with hepatic steatosis and juvenile semilethality.

SAMPLE BUFFER CONSTITUENTS FOR 2DE

Solubilization, disaggregation, denaturation, and reduction of proteins during or right after cell disruption is achieved by the use of chaotropic agents, detergents, reducing agents, buffers, and carrier ampholytes. These buffer constituents must be compatible with IEF, both electrically and chemically.

Chaotropic Agents

In a nondenaturing 2DE-urea, a neutral chaotropic agent, such as urea, can act as a denaturing agent in sample solutions at concentrations typically at 8–9 M. It effectively disrupts noncovalent and ionic bonds between amino acid residues. Urea-containing solutions may not be heated above 37°C to avoid degradation of urea to cyanate. Cyanate ions can react with the amine groups of proteins which can cause carbamoylation and remove the positive charge of the amine, thereby affect the p*I* of the respective proteins. The solubilizing power of urea-containing sample solutions can be dramatically increased by the addition of thiourea. Typically, thiourea is used at 2 M concentration in conjunction with 5–7 M urea.

Detergents

Hydrophobic interactions within a polypeptide chain or between proteins in protein complexes can be disrupted in the presence of detergents. The use of detergents can also increase solubility, especially of membrane proteins. Detergents operate synergistically with chaotropic agents during the solubilization process. This is because they prevent hydrophobic interactions between hydrophobic protein stretches exposed by the chaotropic agents. Detergents consist of a hydrophobic tail and a hydrophilic head moiety, which may be anionic, cationic, zwitterionic, or nonionic. To allow proteins to migrate according to their own charge during IEF, zwitterionic or nonionic detergents are preferred. Although nonionic detergents such as Nonidet P-40 or Triton X-100 can be used, the zwitterionic detergent CHAPS has much better solubilization, particularly for membrane proteins. The selection of the optimal detergent in a solubilization cocktail should take into account how the detergent interacts with high concentrations of urea.

SDS is excellent in its ability to efficiently and rapidly solubilize proteins. Although SDS is incompatible with IEF as an anionic detergent, it can be used in the initial preparation of concentrated protein samples. In these cases, another IEF-compatible detergent must be used in excess to disrupt the binding of SDS to protein. The IEF-compatible detergent: SDS concentration ratio should be at least 8:1 to avoid detrimental effects of SDS in IEF. In addition, its final concentration in the sample solution buffer should be below 0.25%.

Reducing Agents

Reducing agents cleave disulfide bond cross-links within and between protein subunits, thereby promoting protein unfolding and maintaining proteins in their fully reduced states. The compounds mostly used for 2D sample preparation are sulfhydryl reducing agents like DTT. A large excess of sulfhydryl reducing agent will shift the equilibrium of the oxidation/reduction reaction toward the fully reduced protein state. Therefore, sulfhydryl reducing agents are typically used in large excess. Typically, DTT is present in sample preparation cocktails at concentrations ranging from 20 mM to 100 mM to ensure reduction of the protein disulfide bonds. Therefore, it is necessary that DTT be evenly distributed over the whole length of the IPG strip during IEF in order to maintain the reduced state of all cysteines present in the protein. On the other hand, DTT is a weak acid and it can migrate toward the anode. As a consequence, some disulfide bonds will re-form, leading to precipitation of some disulfide rich-proteins. Artificial spots might be generated because of the formation of disulfide bridges. Phosphines such as tributylphosphine offer an alternative to thiols as reducing agents because they are uncharged and reduce cystines stoichiometrically at concentrations as low as 2 mM.

Another approach to avoid protein aggregation and precipitation in basic regions (above pH 8) of IPG gradients is to reduce proteins with tributylphosphine and then irreversibly alkylate with iodoacetamide (Fig. 8.16). Tributylphosphine is a more effective reducing agent than DTT because of it ability to increase protein solubility. This treatment, which is

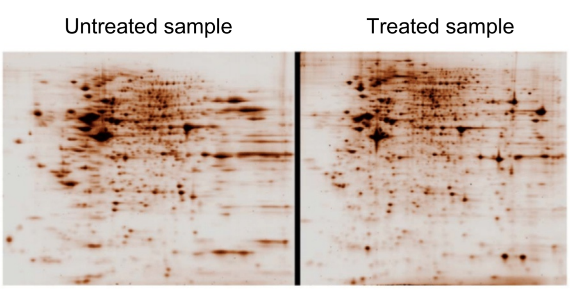


FIG. 8.16 HeLa cell extract separated by 2DE and stained to demonstrate the effect of reduction and alkylation treatment of protein samples (treated sample on right demonstrates better spot resolution than untreated sample on left).

performed after protein solubilization and before IEF, blocks protein sulfhydryls and prevents proteins from aggregating and precipitating due to oxidative cross-linking. This can ensure that proteins remain soluble throughout electrophoresis.

Carrier Ampholytes

Carrier ampholytes are usually included at a concentration of 0.5%–2% (v/v) in sample solutions for IPG strip focusing. They can reduce protein-matrix hydrophobic interactions, which tend to occur at the basic end of the IPG strip and lead to streaking caused by precipitation. Carrier ampholytes can also overcome detrimental effects from salt boundaries and help to compensate for insufficient salt in a sample. This is because even in the presence of detergents, certain samples may have stringent salt requirements to maintain the solubility of some proteins. Although IPG strips are tolerant to salts, salts should be in the sample only if they are absolutely required. During IEF, any salt will be removed, and proteins that need salt for solubility are then subject to precipitation. As such, salts forming strong acids and bases (e.g., NaCl, Na_2HPO_4) should be avoided, because strongly alkaline cationic and strongly acidic anionic boundaries are formed by the migration of the salt's ionic constituents. Salts formed from weak acids and bases such as Tris-acetate and Tris-glycine should be used instead.

Removal of Interfering Contaminants

Sample purity can determine success or failure of any protein analysis. Interfering substances are any contaminants that negatively impact IEF, SDS-PAGE, or both. Commonly seen contaminants include salts, detergents, small compounds, nucleic acids, lipids, and polysaccharides. Therefore, it is important to eliminate contaminants from the original biological sample during sample preparation. Four widely applied precipitation methods are trichloroacetic acid, acetone, chloroform/methanol and ammonium sulfate, and ultrafiltration. Protein precipitation with trichloroacetic acid and acetone as well as ultrafiltration results in efficient sample concentration and desalting. However, independent of precipitation and resolubilization method, protein recoveries in the liquid phase are rarely 100%. All protocols require careful optimization to ensure a true protein representation in the final 2D image.

GEL FREE PROTEOMICS TECHNIQUES: CAPILLARY ELECTROPHORESIS

The Principles of Capillary Electrophoresis

Capillary electrophoresis (CE) is a typical gel-free technique, with the advantage of superior separation efficiency, small sample consumption, short analysis time, and automatability. CE can readily be coupled online with MS. CE separates proteins, with or without prior denaturation, in slab gels or microfluidic channels. Protein separation in CE is based on a variety of properties, including their molecular mass (SDS-capillary gel electrophoresis [CGE]), their isoelectric point (capillary isoelectric focusing [CIEF]), or their electrophoretic mobility (capillary zone electrophoresis [CZE]).

In SDS-CGE, a capillary is filled with gel or viscous solution (to form molecular sieve) and the electroosmotic flow (EOF) is usually suppressed. EOF is a unique driving force in CE that is generated at the wall of a fused-silica capillary by the formation of an electrical double layer associated with the ionized silanol groups. Therefore, the analytes migrate solely based on their electrophoretic mobility. CGE can be accomplished using either a permanently coated gel (e.g., cross-linked polyacrylamide) or a dynamic coated gel (e.g., linear polyacrylamide and cellulose dextran). In the first approach, the gel is prepared and bonded inside the fused silica capillary by polymerization of monomer, whereas in the second approach, the gel is dissolved in aqueous solution to form molecular sieve and used as a separating medium.

With CZE, sample is introduced into a buffer-filled capillary either electrokinetically (with low voltage) or hydrodynamically (with pressure or suction). Both ends of the capillary and electrodes are then placed into a buffer solution that also contains the electrodes, and a high voltage is applied to the system. The applied voltage causes the analytes to migrate through the capillary and past a detector window, where information is collected and stored by an appropriate data acquisition system.

Microchip Electrophoresis

Microchip electrophoresis (ME) is a miniaturized form of conventional CE. In ME, the basic CE configuration has been miniaturized and transferred to a chip format. With ME, it is possible to integrate injection, separation, and detection on a single microchip with typical channel lengths of 5 cm to 15 cm. The separation mechanism for ME is similar to that of CE in that differential migration of analytes is governed by their effective ionic mobilities and the mobility of the EOF under an electric field (Fig. 8.17). In ME, the EOF can be generated on different materials rather than just silica. Compared to conventional CE the advantage to use ME include shorter analysis time, less sample/reagent consumption, and less

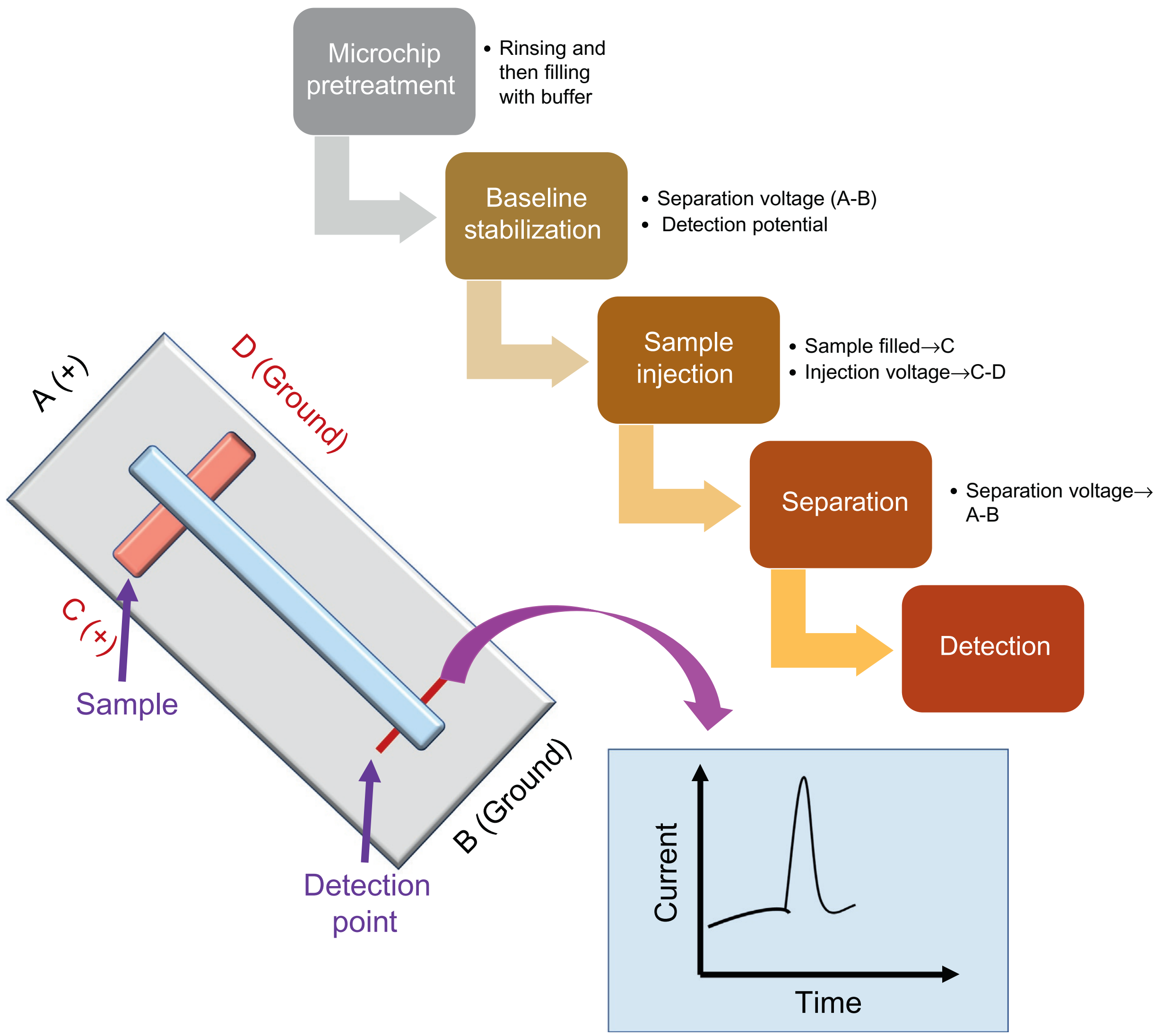


FIG. 8.17 The process of microchip capillary electrophoresis.

waste generation. Furthermore, ME requires lower voltages due to the shorter separation channels, and the associated instrumentation can be made small, leading to portable analytical systems. ME enables fast separation of protein, peptide, and single-cell analysis, which are normally present in small amounts in complex matrices.

GEL FREE PROTEOMICS TECHNIQUES: LIQUID CHROMATOGRAPHY

Liquid chromatography is the fundamental of protein separation science. It can detect molecules at the nanomolar level. Therefore, it is useful in proteomics and genome research. Many kinds of liquid chromatography, such as reversed-phase high performance liquid chromatography (HPLC), affinity HPLC, gel permeation HPLC, ligand-exchange HPLC, and capillary HPLC, have been used in proteomic research. The principles exploited in HPLC basically are the same as those used in common column chromatographic methods such as affinity chromatography, ion-exchange chromatography, or size-exclusion chromatography. The difference is that very high-resolution separations can be achieved quickly and with high sensitivity in HPLC using automated instrumentation.

Reversed-Phase HPLC

Reversed-phase HPLC is the most popular mode of chromatography because of its wide range of applications and the availability of various mobile and stationary phases. Stationary phases mostly comprise nonpolar alkyl hydrocarbons such as C8

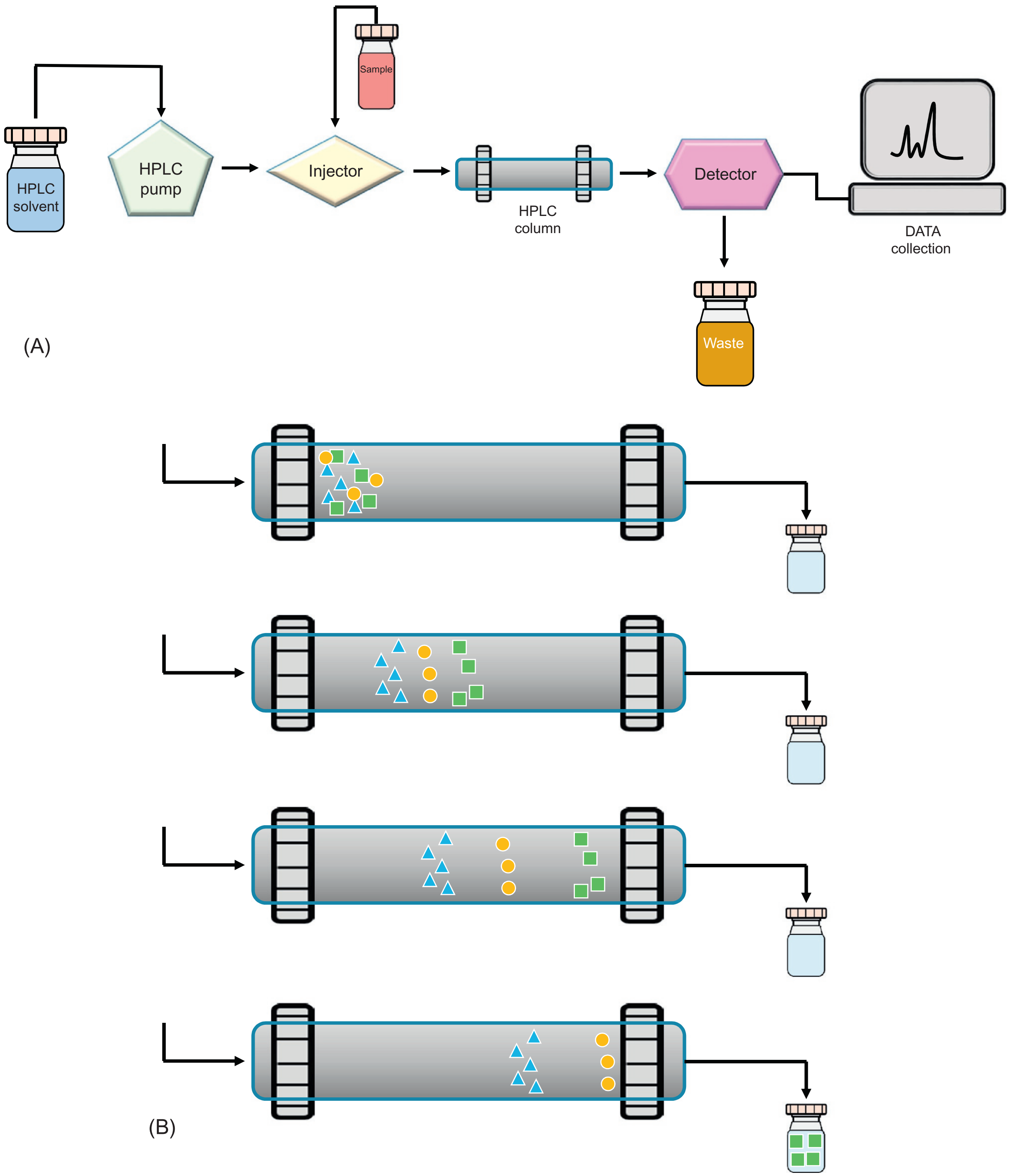


FIG. 8.18 (A) The flow chart of HPLC. (B) The migration of molecules in an HPLC column.

or C18 chains bound to silica or other inert supports (Fig. 8.18). C18 and C8 refer to the alkyl chain length of the bonded phase of the column. The chain length influences hydrophobicity of the sorbent phase and thus increases retention of ligands. C18 is often called the "**traditional reverse phase matrix**" because it has the highest degree of hydrophobicity. The reason why C-18 is more hydrophobic than other reverse phases is because length of the carbon chains are longer. C8 is used when shorter retention times are desired. Lower hydrophobicity means faster retention for nonpolar compounds, hence nonpolar compounds move down the column more readily with C8 than with C18. C8 is preferred over C18 if one is looking for a reverse phase matrix that has a lower degree of hydrophobicity. C18 columns actually can handle more than 60% of the applications in most HPLC labs. The mobile phase is polar, and the elution order is polar followed by less polar, and weakly polar or nonpolar compounds in the end.

Affinity HPLC

Affinity HPLC is a chromatographic method capable of separating biochemical mixtures of highly specific nature. It is possible to design a stationary phase that reversibly binds to a known subset of molecules just by combining affinity chromatography. This method exploits a well-known and well-defined property of analytes that can be used during purification process. The process can be considered as an entrapment, with the target molecule trapped on a stationary phase while the other molecules in solution are not trapped because they lack this property.

Gel-Permeation HPLC

Gel-permeation HPLC works on the principle of sizes of the compounds, which is the same as the size-exclusion chromatography. In this technique, large molecules are eluted first, followed by smaller molecules. Therefore, it is a method of choice for separation of biomolecules such as peptides, proteins, and enzymes. The stationary phase is a porous solid such as glass or silica, or a cross-linked gel that contains pores of appropriate dimensions to affect the separation desired. The separation and isolation of proteins can be done by using Sepharose chromatography followed by MS.

Ligand-Exchange HPLC

Ligand-exchange HPLC is an advanced version of reversed-phase-HPLC where the reversed-phase column is replaced by an ion-exchange column. It has been used widely for the analysis of all inorganic and organic ionic species. In ligand-exchange-HPLC, either anion- or cation-exchange column is used but, nowadays, mixed (anion and cation) columns are also available that improve the separation efficiency.

In cation-exchange chromatography, the stationary phase is usually composed of resins containing sulfonic acid groups or carboxylic acid groups of negative charge, and thus cationic molecules are attracted to the stationary phase by electrostatic interaction. In anion-exchange chromatography, the stationary phase is a resin, generally, containing primary or quaternary amine functional groups of positive charge, and thus these stationary phase groups attract solutes of negative charge.

Capillary Electrochromatography

A hybrid technique of HPLC and CE is capillary electrochromatography (CEC). It combines the high peak efficiency that is characteristic of electrically driven separations with high separation selectivity (Fig. 8.19). CEC can be carried out on wall coated open tubular capillaries or capillaries packed with particulate or monolithic silica or other inorganic materials as well as organic polymers. The chromatographic and electrophoretic mechanisms work simultaneously in CEC, and several combinations are possible.

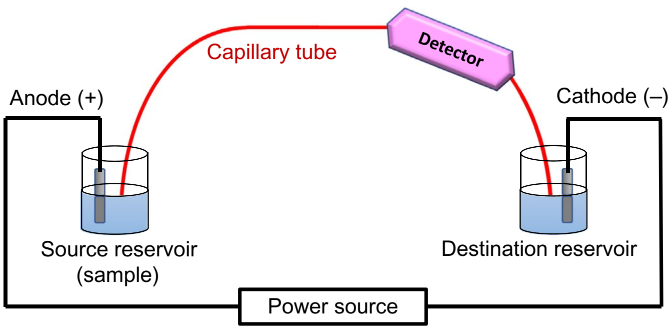


FIG. 8.19 A schematic of capillary electro-chromatography.

Comparison of Various Chromatographic Methods

All chromatographic techniques are important and useful for proteomics analysis. The choice of which technique to use depends on the type and nature of the proteins to be analyzed. For example, gel-permeation HPLC is useful for proteomic separation and identification because of the wide variation in the sizes of proteins. Ligand exchange is useful because proteins have charges, which may be exploited in this kind of chromatography. The comparison of each chromatographic method and their applications are listed in Table 8.4. Among various chromatographic methods used in proteomic analyses the order of application is reversed phase > gel permeation > ligand exchange > affinity. The most popular method

TABLE 8.4 Various Chromatographic Techniques for Proteomic Analyses

Proteomes	Columns	Mobile Phases
Reversed-phase high performance liquid chromatography		
Complex protein mix	C_{18} column	Acetonitrile-TFA
Cytoskeletal proteins & enzymes	PepMap C_{18}	–
Peptides of *Caenorhabditis elegans*	C_{18}	Acetonitrile-formic acid
S-transferase isoenzyme	Vydac 214TP C_4	Acetonitrile and water with TFA
Photosytem II antenna protein	Vydac C_4	Acetonitrile-water-TFA
Proteins of rat adipose cells	Zorbax 300 SB-C3	Acetonitrile-TFA
Complex protein mixture	Zorbax SB-C_{18}	Buffer-acetonitrile-FA Buffer-acetonitrile-TFA
GPI-Aps protein	Zorbax SB-C_{18}	Acetonitrile with acids HFBA (different combinations)
Phosphopeptides of rat kidney inner medullary collecting duct	Picofrit RP column	Acetonitrile-FA
Venoms of snakes	Lichrosphere RP100 C_{18}	Water-acetonitrile-TFA
Affinity high performance liquid chromatography		
Peptides of liver of mice	Magic C_{18} AQ	Water-acetonitrile-FA
Phosphopeptides of *Arabidopsis*	PepMap C_{18}	Water-acetonitrile-FA
Ligand-exchange high performance liquid chromatography		
Proteomic analysis of *E. coli*	218 TP 5415 Vydac C_{18} RP column	Acetonitrile-water-TFA
Pancreatic islet proteome	Polysulfoethyl A column	10 mM Ammonium formate buffer-water and acetonitrile
Membrane proteins of breast cancer MCF7 and BT474 cells	Polysulfoethyl A resin	–
C. elegans peptidomic analysis	Bio-SCX column	Water-acetonitrile-FA
Capillary electro-chromatography		
Protein of *Helicobacter pylori*	ReproSil-Pur C_{18} AQ	–
Tryptic peptide of fish	Phenomenex Jupiter C_{18}	Acetonitrile-TFA-formic acid
Synaptic proteomes of wild type mice	Polysulfoethyl A	Buffers-acetonitrile-10 mM KH_2PO_4
Proteomic analysis of *Escherichia coli*	TSKG3000SWxL	Water-50 mM KH_2PO_4-200 nM NaCl
Vacuum-dried peptides	Jupiter C_{18} RP capillary column	Water with TFA and FA, water and acetonitrile with TFA

FA, formic acid; *HFBA*, heptafluorobutyric acid; *TFA*, trifluoroacetic acid.

used in proteomic analyses is reversed phase-HPLC. This is because that it is well developed. There are many types of reversed-phase stationary phases available, which can be used for analyses of proteomes. Besides, the reversed-phase columns are able to work with a wide range of mobile phases, enhancing the application range of reversed-phase chromatography.

PROTEIN ANALYSIS: MASS SPECTROMETER

After the proteins are separated and quantified, the next step is to identify each protein in the sample. For example, individual spots are cut out of the gel and cleaved into peptides with proteolytic enzymes. These peptides can then be identified by MS. Mass spectrometers exploit the difference in the mass-to-charge (m/z) ratio of ionized atoms or molecules to separate them from each other. The basic operation of a mass spectrometer is to (1) evaporate and ionize molecules in a vacuum, creating gas-phase ions; (2) separate the ions in space and/or time based on their m/z ratios; and (3) measure the quantity of ions with specific m/z ratios. Different MS have been developed depending on the method to ionize proteins for analysis. The two most prominent MS methods for protein analysis are electrospray ionization MS (ESI-MS) and matrix-assisted laser desorption/ionization-time of flight MS (MALDI-TOF MS).

ESI-MS

A solution of macromolecules is sprayed in the form of fine droplets from a glass capillary under the influence of a strong electrical field (Fig. 8.20). The droplets pick up positive charges as they exit the capillary; evaporation of the solvent leaves multiply charged molecules. Individual protein molecules acquire about one positive charge per kilodalton. The MS spectrum of these charged molecules is a series of sharp peaks whose consecutive m/z values differ by the charge and mass of a single proton, leading to a spectrum of m/z ratios for a single protein species (Fig. 8.21). Computer analysis can convert these data into a single spectrum that has a peak at the correct protein mass. ESI-MS recently is evolved to the so-called nanospray ionization, while "tandem" MS is rapidly replaced by "cascades" of mass spectrometers using more sophisticated and sensitive detectors (such as orbit traps), and accommodating other intermediate "sequencing" instruments, like electron-transfer dissociation.

MALDI-TOF

In MALDI-TIF procedure, a peptide is placed on a matrix, which causes the peptide to form crystals (Fig. 8.22). A laser pulse is used to excite the chemical matrix, creating a microplasma that transfers the energy to protein molecules in the sample, ionizing them and ejecting them into the gas phase. An increase in voltage at matrix is used to shoot the ions toward

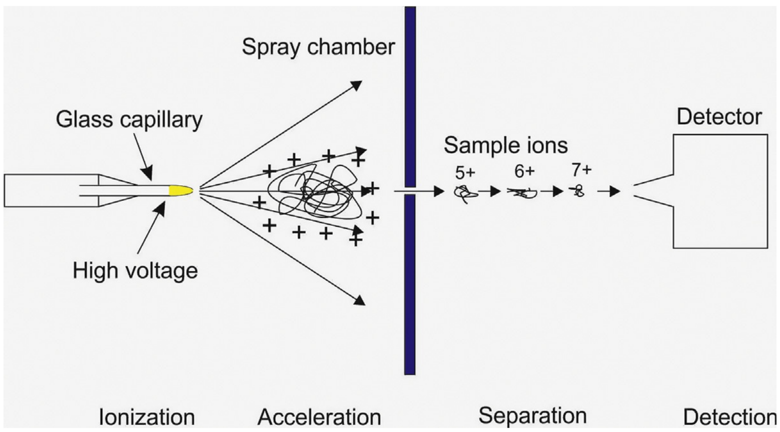


FIG. 8.20 The steps of electrospray ionization mass spectrometry (ESI-MS).

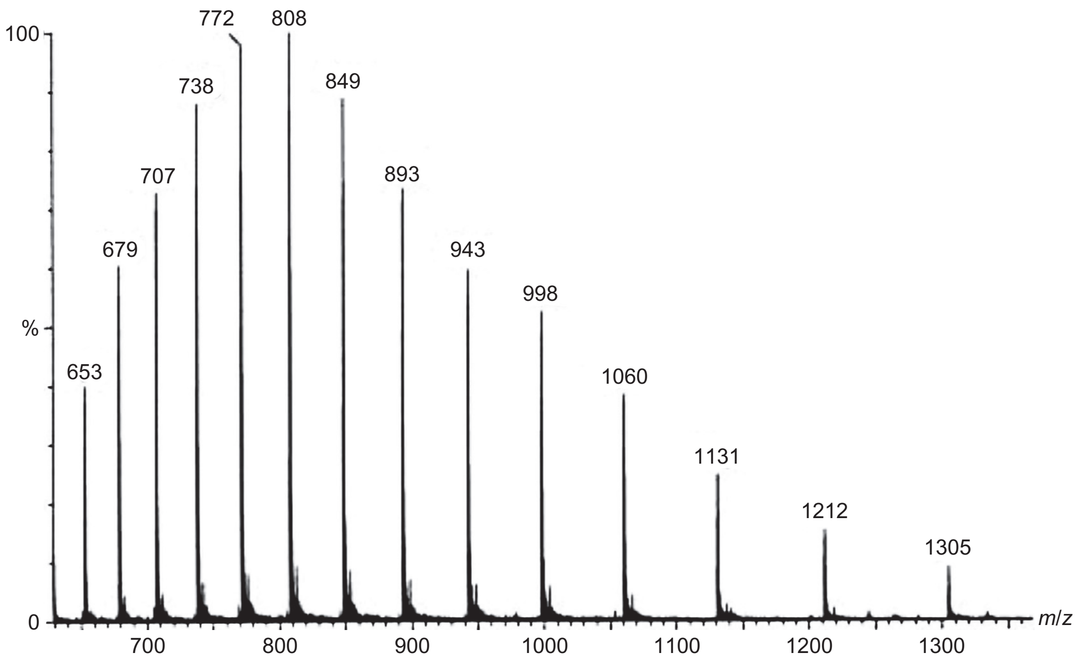


FIG. 8.21 A typical protein analysis via electrospray ionization mass spectrometry. This analysis represents the cluster ions of a highly purified horse myoglobin protein standard solution (5 μmol/L in 50% acetonitrile solution containing 0.2% formic acid).

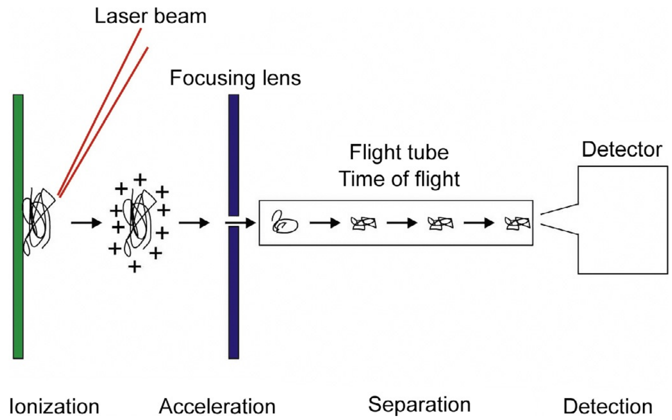


FIG. 8.22 MALDI-TOF mass spectrometry mechanism.

a detector. Protein molecules that have picked up a single proton can be selected by the MS for mass analysis. This is because the time it takes an ion to reach the detector depends on its mass. The higher the mass, the longer the time of flight of the ion. In a MALDI-TOF MS, the ion can also be deflected with an electrostatic reflector that also focuses the ion beam. Thus, one can determine the mass of the ions reaching a second detector with high precision, and these masses can reveal the exact chemical composition of the peptides. As such, MALDI-TOF MS is very sensitive and accurate for determining amino acid sequences.